

Hankel Dynamic Mode Decomposition with control for temperature diffusion monitoring

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1. Introduction

Standard Dynamic Mode Decomposition (DMD) is a powerful tool for discovering coherent structures in high-dimensional data. However, when the observation dimension is low (e.g., a single sensor probe) or when the system exhibits standing waves, standard DMD may fail to capture the full dynamic.

Hankel DMD overcomes this by using **time-delay embeddings**. Instead of analyzing a single snapshot in time, time-delay embeddings are used to augment the dimension of the data. This enriches the state space, allowing us to capture dynamics that are unobservable in the original low-dimensional measurement.

In this application case we want to achieve a model that can predict the temperature raise of 2 switches based on the power loss, Time-delay embeddings of 2 temperature probes $[T1_n, T2_n]$ and power inputs, $[P1_n, P2_n]$ are stacked and phase lagged to run the **DMDc (Dynamic Mode Decomposition with control)** algorithm, proposed by (Proctor, Brunton, and Kutz 2014), and produce a state transition matrix M.

Additionally, an Online DMD algorithm was implemented as a monitoring tool as described in (Zhang et al. 2017)

2. Mathematical Formulation

The Hankel Matrix

Given a time series of data measurements $\mathbf{x} = [x_k, x_{k+2}, \dots, x_N]$, where each x_k can be a scalar or a vector, we construct a Hankel matrix $\mathcal{H}$.

In this application case $x_k = [T1_k, T2_k]$

If we choose a **stacking depth** (or lag embedding dimension) d , the Hankel matrix arranges the data such that each column contains a trajectory of length d :

$$\mathcal{H}_{\mathcal{X}} = \begin{bmatrix} \mathbf{x}_k & \mathbf{x}_{k+1} & \dots & \mathbf{x}_N \\ \mathbf{x}_{k-1} & \mathbf{x}_k & \dots & \mathbf{x}_{N-1} \\ \vdots & \vdots & \ddots & \vdots \\ \mathbf{x}_{k-d} & \mathbf{x}_{k-d+1} & \dots & \mathbf{x}_{N-d+1} \end{bmatrix}_{d \times N-d}$$

The Omega Matrix

For DMDc we define a input snapshot matrix $U_N = [u_k, u_{k+2}, \dots, u_N]$ where each u_k can be a scalar or a vector. In this application $u_k = [P1_k, P2_k]$ with a Hankel matrix $\mathcal{H}_{\mathcal{U}}$.

To obtain the augmented matrix Ω we construct

$$\Omega = \begin{bmatrix} \mathcal{H}_{\mathcal{X}} \\ \mathcal{H}_{\mathcal{U}} \end{bmatrix}$$

For the temperature rise modelling in this application 2 experiments $[A, B]$ are conducted with $P_1 = P, P_2 = 0$ and $P_1 = 0, P_2 = P$, in order to capture the effects of each source the Hankel matrix $\Omega = [\Omega_{\mathcal{A}} \Omega_{\mathcal{B}}]$ is assembled.

In DMD we want to compute a transition Matrix M such as $X_N = \Omega_{N-1}M$, since in this problem we have inputs P_1, P_2 the problem is formulated using DMDc $Z_N = Z_{N-1}A + U_NB$

The DMDc Algorithm

Once we have Ω , we proceed with the standard DMD algorithm:

- Create two time-shifted matrices from $\mathcal{H}$:
 - $\Omega_{N-1} = \Omega[k : N - 1]$ (The past)
 - $X_N = \mathcal{H}_{\mathcal{X}}[k + 1 : N]$ (The future)
- Compute the Singular Value Decomposition (SVD) of Ω_{N-1} : $\Omega_{N-1} \approx U \Sigma V^*$.
- Identify the most energetic modes in Σ .
- Truncate matrices to new dimension e.g m dominant modes $\tilde{U} = U[:, : m]$ $\tilde{\Sigma} = \Sigma[:, : m]$ $\tilde{V}^* = V^*[:, : m]$
- Split $\tilde{U}$ into the state $U_x = \tilde{U}[:, d \times states, :]$ and input components $U_u = \tilde{U}[d \times states :, :]$
- Project X_N into the reduced subspace $U_x \tilde{X}_N = U_x^* X_N$
- Compute the operator $\tilde{M}$ that maps Ω_{N-1} to the future X_N :

$$\tilde{M} = \tilde{U}^* \tilde{X}_N \tilde{V} \tilde{\Sigma}^{-1}$$

- Compute $\tilde{A} = \tilde{M} \tilde{U}_x^* \tilde{U}_x$; $\tilde{B} = \tilde{M} \tilde{U}_u^*$ and state $\tilde{Z}_{N-1} = \tilde{U}_x^* X_{N-1}$
- Compute the eigenvalues λ and eigenvectors W of $\tilde{A}$.
- Diagonalize $\tilde{A}_d = W^{-1} \tilde{A} W$; $\tilde{B}_d = W^{-1} \tilde{B}$ and the new eigenbasis state $\tilde{Z}_d = W^{-1} \tilde{Z}$

3. Online DMD

For the real time monitoring of time constants in time varying systems, such as in the case of thermal wear/degradation, the online DMD algorithm is useful to update the $\tilde{A}_d$ matrix at each time step, while using the offline DMD results as a healthy baseline.

Formulation & Algorithm

$$X_k = G_k X_{k-1}$$

where the matrix G_k is the system matrix at time k $G = [AB]$

G_k is updated based on a matrix P_k which is the autocorrelation matrix of all data to time k.

$$P_k = (\sum_{i=1}^k X_i X_i^*)^{-1}$$

Initialization

- $G_0 = [A_d B_d]$ obtained in the offline DMD
- $P_0 = \hat{\Omega}_d \hat{\Omega}_d^*$ where

$$\tilde{\Omega}_d = \begin{bmatrix} W^{-1} \mathcal{H}_x \\ \mathcal{H}_u \end{bmatrix}$$

Update

- Compute the gain γ

$$\gamma = \frac{P_k X_k}{\rho + X_k^* P_k X_k}$$

where ρ is a forgetting factor and $X = \begin{bmatrix} \tilde{Z}_d \\ u \end{bmatrix}$

- Update P

$$P_{k+1} = \frac{1}{\rho} (P_k - \gamma (P_k X_k)^*)$$

- Update G

$$G_{k+1} = G_k + (X_{k+1} - G_k X_k) \gamma^*$$

The eigenvectors and eigenvalues of matrix A can be extracted from G to monitor the system dynamics and drift.

References

- Proctor, Joshua L., Steven L. Brunton, and J. Nathan Kutz. 2014. “Dynamic Mode Decomposition with Control.” arXiv. <https://doi.org/10.48550/arXiv.1409.6358>.
- Zhang, Hao, Clarence W. Rowley, Eric A. Deem, and Louis N. Cattafesta. 2017. “Online Dynamic Mode Decomposition for Time-Varying Systems.” arXiv. <https://doi.org/10.48550/arXiv.1707.02876>.